Please write clearly in block capitals. Centre number | Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE

Paper 2 (Generated practice paper - Set 10)

Assessed: The living environment, Biological resources, Sustainability, Research methods

Time allowed: 3 hours

Materials

For this paper you may use:
- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
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11	
TOTAL	

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Answer all questions in the spaces provided.

0 1 The Rio Bravo Conservation and Management Area in Belize exploits mahogany under a sustainable management plan while conserving a rainforest of over 200 tree, 400 bird and 70 mammal species.

0 1 . 1 Describe **five** features of the timber harvesting at Rio Bravo that make it sustainable.

[5 marks]

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0 1 . 2 Explain why involving local communities is essential to the success of the reserve.

[3 marks]

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0 1 . 3 Explain why selective logging is better for wildlife than clear-felling, and give **one** disadvantage.

[2 marks]

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0 2 Table 1 shows characteristics introduced into commercial crops from crop wild relatives (CWRs).

Table 1

Crop	Wild relative source	Characteristic introduced
Sugar cane	Wild sugar cane, Indonesia	Resistance to mosaic virus
Tomato	Single wild species	Resistance to nine major diseases
Rice / barley	Wild salt-marsh varieties	Salt tolerance
Oil palm	Wild palms, central Africa	25% higher yield
Wheat	Wild wheat	Better mycorrhizal associations / phosphate uptake

0 2 . 1 Explain why commercial crop varieties often lack the characteristics shown in Table 1.

[3 marks]

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0 2 . 2 Explain what is meant by a **Vavilov centre** (centre of diversity) and why these areas are conservation priorities.

[3 marks]

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0 2 . 3 Explain why a large population does not necessarily have a large gene pool, and why a small gene pool is a problem.

[3 marks]

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0 2 . 4 Give **one** wild plant species with potential as a new commercial food crop and the characteristic that makes it valuable.

[1 mark]

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0 3 A conservation team is planning to protect a population of an endangered antelope in a reserve.

0 3 . 1 Explain why each of the following features of the population should be measured before a conservation strategy is decided.

[8 marks]

Population size .....

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Distribution .....

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Survival rate .....

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Age structure .....

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0 3 . 2 Describe how the population size of the antelope could be estimated.

[2 marks]

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0 4 Table 1 shows food conversion ratios (FCR) for livestock: the mass of feed needed to produce 1 kg of live mass gain.

Table 1

Livestock	FCR / kg feed per kg gain
Farmed salmon	1.2
Chicken	1.8
Pork	3.0
Beef (feedlot)	7.0
Sheep	8.0

0 4 . 1 Use Table 1 to explain why intensive poultry and fish production convert feed to food more efficiently than beef.

[3 marks]

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0 4 . 2 Explain **three** ways farmers reduce food chain energy losses in livestock production.

[3 marks]

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0 4 . 3 Explain why growing crops for direct human consumption usually produces more food per hectare than rearing livestock, and describe **one** situation where livestock is the better option.

[4 marks]

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0 5 . 1 Which feature of the Earth prevents most atmospheric gases from escaping into space? Tick **one** box.

[1 mark]

A Its mass ☐

B Its magnetic field ☐

C Its distance from the Sun ☐

D Its speed of rotation ☐

0 5 . 2 Explain how the speed of rotation of the Earth and the tilt of its axis affect the conditions for life.

[2 marks]

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0 5 . 3 Explain why the presence of liquid water on Earth depends on **both** the distance from the Sun **and** atmospheric pressure.

[2 marks]

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0 6 . 1 Explain why natural systems regulated by negative feedback rarely reach tipping points, and give an example of a natural tipping point in the geological past.

[3 marks]

0 6 . 2 Explain why diverse systems are more resistant to change than simple systems, using a natural and a human example.

[4 marks]

0 6 . 3 Give **three** positive feedback mechanisms by which human-induced warming may become self-sustaining.

[3 marks]

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0 7 . 1 Explain why carnivorous species such as salmon dominate aquaculture in MEDCs while herbivores such as carp and tilapia dominate in LEDCs.

[3 marks]

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0 7 . 2 Describe how all-female rainbow trout are produced and explain why this is done.

[3 marks]

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0 7 . 3 Explain how each of the following abiotic factors is controlled in an intensive salmon farm.

[4 marks]

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Dissolved oxygen .....

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Water flow .....

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Day length .....

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Temperature .....

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0 8 . 1 Explain how each of the following agricultural changes affects wildlife habitats.

[6 marks]

Drainage of wet grassland .....

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Application of fertiliser to a species-rich meadow .....

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Removal of hedgerows to enlarge fields .....

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0 8 . 2 Explain why some farmland habitats are valuable for wildlife conservation.

[2 marks]

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0 8 . 3 Describe how agriculture contributes to climate change.

[2 marks]

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0 9 . 1 Describe the ecological features of mangroves and the adaptations of mangrove trees.

[3 marks]

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0 9 . 2 Explain **four** reasons why mangroves are important.

[4 marks]

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0 9 . 3 Describe the main threats to mangroves and the conservation efforts used.

[3 marks]

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1 0 . 1 Evaluate the extent to which the principles of the circular economy can be applied to **three** of the following: water supplies, mineral supplies, waste management, pollution control, energy supplies, food production.

[9 marks]

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Extra space

1 0 . 2 State what is meant by **industrial symbiosis**.

[1 mark]

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Write an essay on one of the following topics.

1 1 . 1 Discuss the causes and consequences of deforestation and evaluate the extent to which forests can be exploited sustainably.

[25 marks]

OR

1 1 . 2 Using named examples, discuss how the ecological features of a habitat determine its importance, the threats it faces and the conservation strategies that are appropriate.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

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END OF QUESTIONS